FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2023

CONCEPT RECAPITULATION TEST – IV

PAPER –2

TEST DATE: 26-04-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 – 04, 19 – 22, 37 – 40)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 23 – 28, 41 – 46)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 18, 29 – 36, 47 – 54)*: This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

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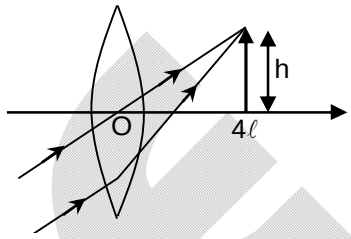
Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. A thin converging lens L_1 forms a real image of an object located far away from the lens as shown in the figure. The image is located a distance 4ℓ and has height h. A diverging lens of focal length ℓ is placed at a distance 2ℓ from lens L_1. Another converging lens of focal length 2ℓ is placed 3ℓ from lens L_1. The height of final image this formed is (Both diverging and converging lenses are placed at right side of L_1)



(A) h (B) $\frac{h}{2}$

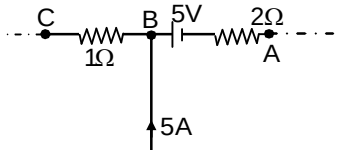
(C) 4h (D) 2h

2. A meteorite approaching a planet of mass M (in the straight line passing through the centre of the planet) collides with an automatic space station orbiting the planet in a circular trajectory of radius R. The mass of the station is ten times as large as the mass of the meteorite. As a result of the collision, the meteorite sticks in the station which goes over to a new orbit with the minimum distance R/2 from the planet. Speed of the meteorite just before it collides with the planet is:

(A) $\sqrt{\frac{58GM}{R}}$ (B) $\sqrt{\frac{38GM}{R}}$

(C) $\sqrt{\frac{28GM}{R}}$ (D) $\sqrt{\frac{18GM}{R}}$

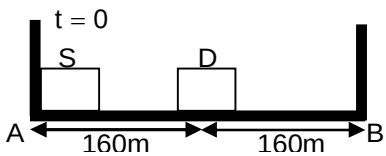
3. If an ideal cell of emf 5 volt shown in the figure gives a power of 10 W, find the powers consumed by the resistors 2Ω and 1Ω.

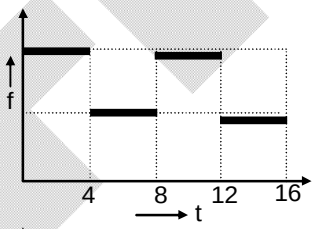


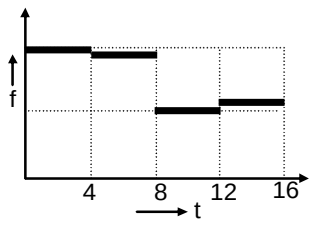
(A) 2W, 18W (B) 8W, 49W

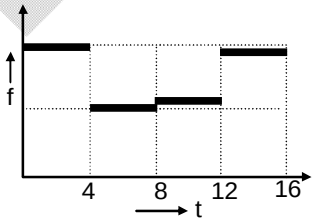
(C) 8W, 18W (D) 2W, 49W

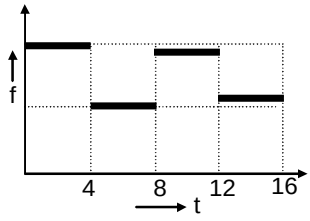
4. A small source S of mass m starts moving with velocity 40 m/s towards a small detector D of same mass m on a frictionless surface AB as shown in the figure. The frequency received by the detector as a function of time in the time interval $0 \le t \le 16$ s is: [Assume all collision to be elastic]. Take $v_s = 340$ m/s.



(A)


(B)


(C)


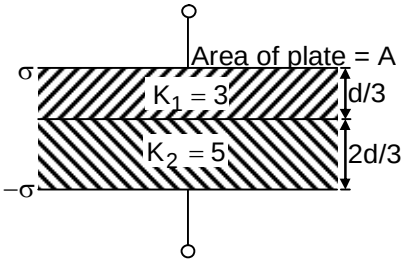
(D)


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Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. In the figure shown, a charged capacitor is filled with two dielectric slabs. Which of the following is/are correct?



(A) the ratio of energy density in 1st dielectric to second dielectric is $\frac{5}{3}$

(B) the ration of energy density in 1st dielectric to second dielectric $\frac{1}{1}$

(C) total induced surface charge on the interface of the two dielectric is $\frac{2\sigma}{15}$

(D) total induced surface charge in the interface of the two dielectric is $-\frac{2\sigma}{15}$

6. If the two slits of double slit experiment were moved symmetrically apart with small relative velocity v and the distance between screen and mid-point of slits is fixed and equal to D. Consider a point P on the screen at a distance x from central maxima then $(x <<< D)$:

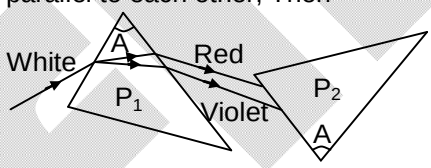
(A) Rate of change of number of fringes between central maxima and point P changes with respect to time is $\frac{xv}{\lambda D}$

(B) number of fringes contained between central maxima and point P increases with time

(C) fringe width decreases as time passes

(D) fringe width increases as time passes

7. P_1 and P_2 are identical prisms arranged as shown in figure. A ray of white light incident on face of P_1 undergoes dispersion and falls on one face of P_2. The facing surfaces of the prisms are parallel to each other, Then



(A) Light emerging from P_2 will be white

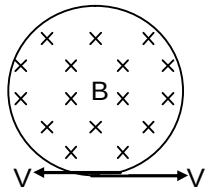
(B) In the light emerging from P_2 dispersion will be greater

(C) the direction of light emerging from P_2 will be parallel to the direction of ray incident on P_1

(D) the ray emerging from P_2 will be white even if prisms P_1 and P_2 have identical geometrical but different material

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8. A circular conducting loop of radius r_0 and having resistance per unit length λ as shown in the figure is placed in a magnetic field B which is constant in space and time. The ends of the loop are crossed and pulled in opposite directions with a velocity v such that the loop always remains circular and the radius of the loop goes on decreasing, then



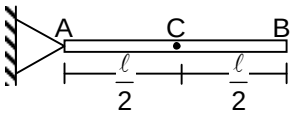
(A) Radius of the loop changes with t as $r = r_0 - (vt/\pi)$

(B) EMF induced in the loop as a function of time is $e = 2Bv[r_0 - vt/\pi]$

(C) Current induced in the loop is $I = \dfrac{Bv}{2\pi\lambda}$

(D) Current induced in the loop is $I = \dfrac{Bv}{\pi\lambda}$

9. A thin uniform rod of mass m and length l is free to rotate about a horizontal axis through its one end. The rod is released from horizontal position as shown at $t = 0$. Choose correct option(s) for $t = 0$.



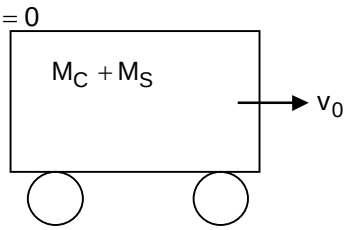
(A) Acceleration of center of mass of part CB is $\dfrac{9g}{8}$

(B) Angular acceleration of rod is $\dfrac{3g}{2l}$

(C) Force exerted by AC on CB is $\dfrac{mg}{16}$

(D) Acceleration of center of mass of AC is $\dfrac{3g}{4}$

10. A railway carriage of mass M_C filled with sand of mass M_S moves along the rails. The carriage is given an impulse and it starts with a velocity v_0. At the same time it is observed that the sand starts leaking through a hole at the bottom of the carriage at a constant mass rate λ. Chose the correct alternative(s). (Neglect the friction along the rails.)



(A) The distance at which the carriage becomes empty is $v_0 \dfrac{M_S}{\lambda}$.

(B) The velocity attained by the carriage at the time, when the carriage becomes empty is v_0.

(C) The time at which the carriage becomes empty is $\dfrac{M_S}{\lambda}$.

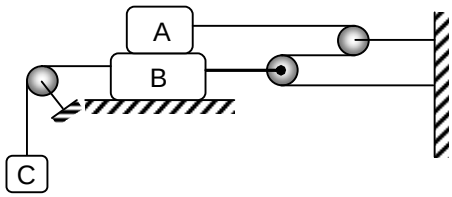
(D) None of these is correct.

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Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

11. Block A of weight 500 N and block B of weight 700 N are connected by rope pulley system as shown. The largest weight C that can be suspended without moving block A and B is W. The coefficient of friction for all plane surfaces of contact is 0.3. The pulleys are ideal. Find $\frac{W}{90}$.

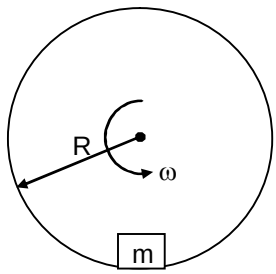


12. Position (in m) of a particle moving on a straight line varies with time (in sec) as $x = \frac{t^3}{3} - 3t^2 + 8t + 4(m)$. Consider the motion of the particle from $t = 0$ to $t = 5$ sec. S_1 is the total distance travelled and S_2 is the distance travelled during retardation. If $s_1 / s_2 = \frac{(3\alpha + 2)}{11}$, then find α.

13. Consider two infinite large parallel current carrying sheet. Current per unit width in both sheets is $\frac{1}{\sqrt{\pi}}$ A / m. If direction of current in both sheets is same and force per unit area on each sheet is $n \times 10^{-7}$ N/m^2. Find n.

14. Two blocks each of mass m are joined together using an ideal spring of force constant K and natural length ℓ_0. The blocks are touching each other when the system is released from rest on a rough horizontal surface. Both the blocks come to rest simultaneously when the extension in the spring is $\frac{\ell_0}{4}$. The coefficient of friction between each block and the surface (assuming it to be same between any of the blocks and the surface) is $\frac{3K\ell_0}{nmg}$. Find n.

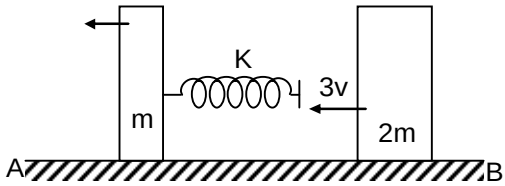
15. A cylinder of radius R is rotating about its horizontal axis with constant $\omega = \sqrt{\frac{5g}{R}}$. A block of mass is kept on the inner surface of the cylinder. Block is moving in vertical circular motion without slipping. Co-efficient of friction between block and surface of cylinder is μ. If minimum value of μ for complete vertical circular motion of block is $\frac{2\sqrt{6}}{3x}$ then find 'x'.



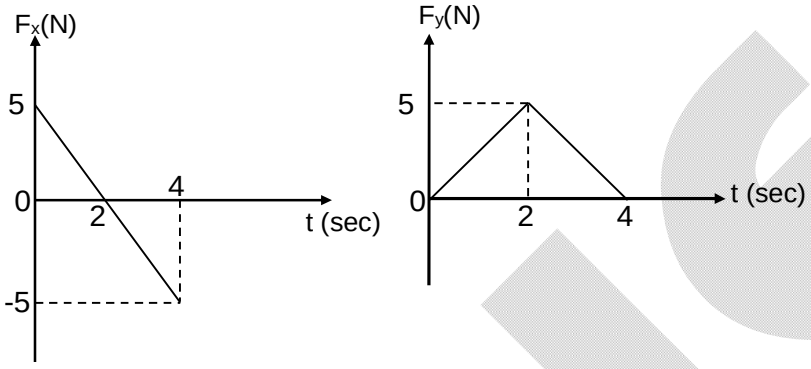
16. A gas consisting of rigid di-atomic molecules (degree of freedom $= 5$) at pressure $P_0 = 10^5 N/m^2$ and temperature 273 K was compressed adiabatically 5 times. The mean kinetic energy of rotating molecules in final state is $n \times 10^{-21}$ J. Find value of n to nearest integer. $\left(K = 1.38 \times 10^{-23}, (5)^{2/5} = 1.90\right)$.

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17. AB is a long frictionless horizontal surface. One end of an ideal spring of spring constant K is attached to a block of mass m. Which is being moved left with constant velocity v, and the other end is free. Another block is mass 2m is given a velocity 3v towards the spring. Magnitude of work done by external agent in moving m with constant velocity v in long time is β times mv^2. Find the value of β.



18. The particle of mass 1 kg is acted upon by a force $\vec{F} = F_x\hat{i} + F_y\hat{j}$ for time $t = 4$ sec starting from $t = 0$. The speed of the particle at $t = 4$ sec., if its initial velocity is $\vec{u} = -\hat{i} + \hat{j}$ and the variation of applied force with time is as shown in the graphs, is $\sqrt{61x}$ m/s, find x.



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Chemistry

PART – II

Section – A (Maximum Marks: 12)

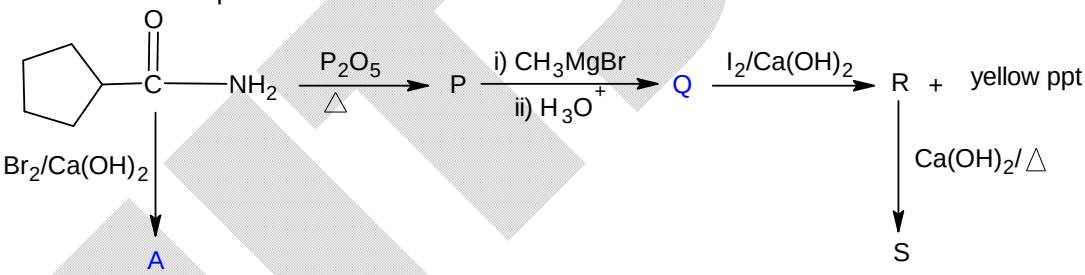
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

19. Given reagents are HCl, NaOH, $ZnCl_2$, Na_2CO_3, NH_4Cl and Zn. Which of these reagent(s) can intensify hydrolysis of $FeCl_3$ when added to its solution?
(A) NaOH, Na_2CO_3
(B) NaOH, NH_4Cl, HCl
(C) Na_2CO_3, NaOH, HCl
(D) NH_4Cl, HCl, $ZnCl_2$

20. Co^{2+} (aq.) + SCN^- (aq.) $\rightarrow$ Complex (X).
Ni^{2+} (aq.) + Dimethylglyoxime $\rightarrow$ Complex (Y).
The coordination number of cobalt and nickel in complexes X and Y are four.
The IUPAC names of the complexes (X) and (Y) are respectively :
(A) tetrathiocyanato-S-cobalt(II) and bis(dimethylglyoximate) nickel(II).
(B) tetrathiocyanato-S-cobaltate (II) and bis(dimethylglyoximato)nickel (II).
(C) tetrathiocyanato-S-cobaltate (II) and bis(dimethylglyoximato)nickelate(II).
(D) tetrathiocyanato-S-cobaltate(III) and bis(dimethylglyoximato)nickel(II).

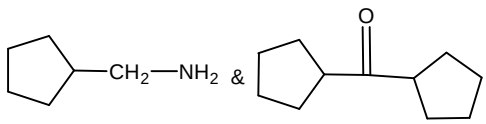
21. Average X –Y bond enthalpy in XY_n is 309.2 kJ/mol. The value of standard enthalpy of formation of XY_n(g), X(g) and Y(g) are –1100, 275 and 80 kJ/mol respectively. XY_n is (Assume X is forming n single bond with Y)
(A) XY_6
(B) XY_4
(C) XY_2
(D) XY_5

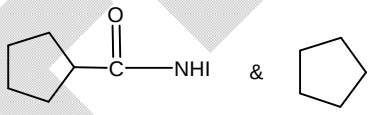
22. In the reaction sequence



A & S respectively are

(A) 

(B) 

(C) 

(D) None of these

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Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. An organic compound (A) having the molecular formula $C_4H_6O_3$ upon heating gives (B) C_3H_6O.

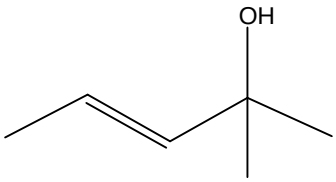
$$(B) \xrightarrow[OH^-]{CH_3CHO} (C) \xrightarrow[(ii)\ NH_4Cl\ (aq)]{(i)\ CH_3Li} (D)\ \text{(major)}$$

$$(D) \xrightarrow[850^oC]{\text{Conc. } H_2SO_4} (E) \xrightarrow{H_2/Ni} (F) \xrightarrow{Br_2/\ h\nu} (G)\ \text{(major)}$$

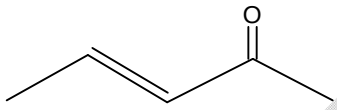
mono bromination

Choose the correct options:

(A) Compound (D) can be :

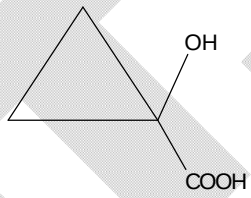


(B) Compound (C) is

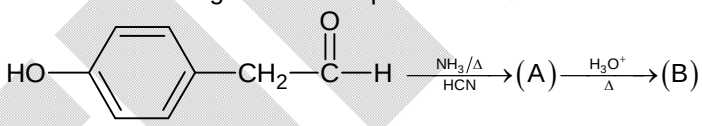


(C) (G) is optically inactive due to external compensation.

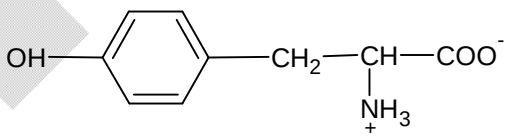
(D) Compound (A) can be



24. From the following reaction sequence:



(A) Compound B is



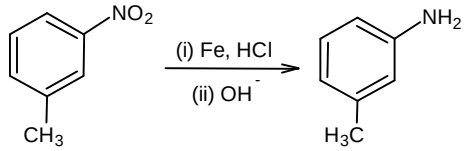
(B) B is racemic mixture

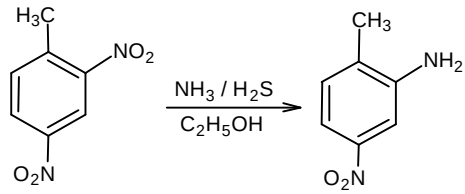
(C) The acidic group present in the compound (B) are - COOH and - OH

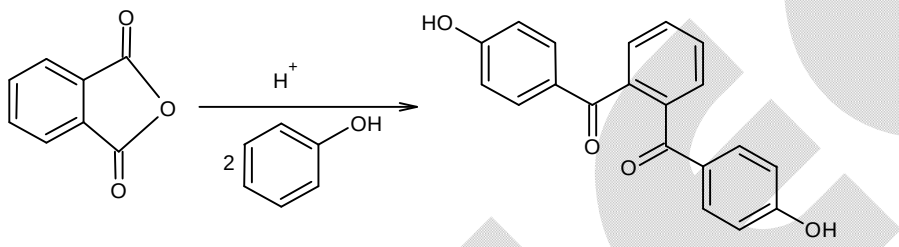
(D) None of the above.

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25. Which of the following reactions can occur?

(A) 

(B) 

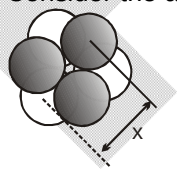
(C) 

(D) None of these

26. One mole of an ideal gas (not necessarily mono atomic) is subjected to the following sequence of steps:
(i) It is heated at constant volume from 300 K to 400 K.
(ii) It is expanded freely into vacuum to triple its volume.
(iii) It is cooled reversibly at constant P to 300 K.
(A) ΔU for the overall process = 0.
(B) ΔH for the overall process = 0.
(C) Total work done is 831.4 joule mol^{-1}.
(D) Total heat content is 831.4 joule mol^{-1}.

27. Select the correct statement(s) in the following
(A) Carbon has pronounced ability to form $P\pi - P\pi$ multiple bonds to itself and to other elements like O and N.
(B) In diamond each carbon atom is linked tetrahedrally to four other carbon atoms by sp^3 hybridisation.
(C) Graphite has planar hexagonal layers of carbon atoms held together by weak Vander Waal's forces
(D) Out of CO_2, SiO_2, SnO_2 and PbO_2. CO_2 , SiO_2 are acidic, SnO_2 is amphoteric and PbO_2 is an oxidizing agent.

28. Consider the arrangement of 6 spheres of radius r



Correct statements are :
(A) Separation between two non touching spheres is $x = 2\sqrt{2}\ r$
(B) Void formed by spheres is found in HCP.
(C) Void formed by spheres is found in FCC
(D) Void formed by spheres is found in BCC

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Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. The sum of E^0 (to the nearest integer) for the following reactions at 298 K is:

$Ag(NH_3)_2^+ + e^- \rightleftharpoons Ag + 2NH_3$

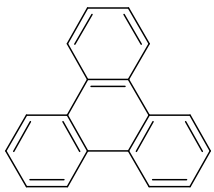
$Ag(CN)_2^- + e^- \rightleftharpoons Ag + 2CN^-$

Given : $E^0_{Ag^+/Ag} = 0.8$ V

K_d^0 for $Ag(NH_3)_2^+ = 10^{-8}$

K_d^0 for $Ag(CN)_2^- = 10^{-19.072}$

30. Number of non charged resonating structures of tryphenylene are



31. In the following reactions:

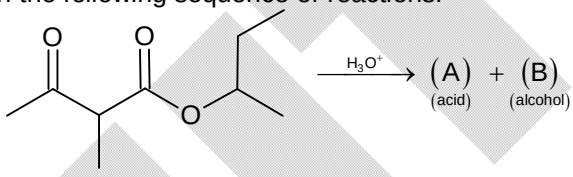
$CaOCl_2 \xrightarrow{\Delta} a\ Ca(ClO_3)_2 + b\ CaCl_2$

$NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + c\ HCl$

$Na_2S_2O_3 + dHCl \longrightarrow e\ NaCl + H_2O + S\downarrow + SO_2$

What will be the value of (a + b + c) – (d + e) in the balanced chemical reactions?

32. In the following sequence of reactions:



(Optically pure compound)

$(A) \xrightarrow{\Delta} (C) + (D)\uparrow$

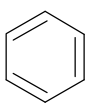
$(C) \xrightarrow{Br_2/h\nu} (E)$ mono brominated products.

What will be the sum of total number of enantiomeric pairs in A, B, C, D and E.

33. The total number of localized lone-pairs of electrons in 1 molecule of Melamine are?

34. Among PbS, CuS, HgS, MnS, Ag_2S, NiS, CoS, ZnS &SnS_2, the total number of black coloured sulphide are:

35. How many 3° amine structures are possible for molecular formula $C_6H_{14}N$?
(Ignore stereo isomers)

36. 

$\xrightarrow[(ii)\ Ph-CH_2-CN;\Delta]{(i)\ HNO_2(conc.)} X,$

Compound 'X' is having no. of pairs of πe^-

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Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. An urn contains three white, six red and four black balls. Two balls are selected at random. What is the probability that one ball is red and other is white? Given that they are of different color.

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{1}{3}$ (D) None of these

38. Let A, B, C are subsets of $X=\{1,2,3,....,n\}$ such that $A\cap B\cap C=\phi, A\cap B\neq\phi, C\cap B\neq\phi$. The number of triplets A, B, C is

(A) $7^n-2\cdot6^n$ (B) $7^n-2\cdot5^n$

(C) $6^n-2\cdot5^n$ (D) None of these

39. Let $f(x)=ax^2+bx+c, a\neq0, a,b,c\in I$. Suppose that $f(1)=0, 50<f(7)<60$ and $70<f(8)<80$. If $13k<f(10)<13(k+1)$ for some integer k, then value of k is

(A) 5 (B) 8

(C) 10 (D) None of these

40. Let $f(x)=x^4+ax^3+bx^2+cx+d$ be a polynomial with real coefficients and real zeroes. If $|f(i)|=1$, (where $i=\sqrt{-1}$), then a + b + c + d is equal to

(A) -1 (B) 1

(C) 0 (D) can't be determined

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. AB and CD are two equal and parallel chords of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Tangents of the ellipse at A and B intersect at P; and tangents at C and D intersects at Q. The line PQ

(A) passes through the origin (B) is bisected at the origin

(C) cannot pass through the origin (D) is not bisected at the origin

42. If the equation $ax^2-6xy+y^2+bx+cy+d=0$ represents pair of lines whose slopes are m and m^2, then value of a is /are

(A) a = - 8 (B) a = 8

(C) a = 27 (D) a = -27

43. If two tangents can be drawn to the different branches of hyperbola $\frac{x^2}{1}-\frac{y^2}{4}=1$ from the point (α,α^2), then

(A) $\alpha\in(-2,0)$ (B) $\alpha\in(0,2)$

(C) $\alpha\in(-\infty,-2)$ (D) $\alpha\in(2,\infty)$

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44. If r_1, r_2, r_3 are the radii of the escribed circles of a triangle ABC and r is the radius of its incircle then the root(s) of the equation $x^2 - r(r_1r_2 + r_2r_3 + r_3r_1)x + r_1r_2r_3 - 1 = 0$ is/are

(A) 1 (B) $r_1 + r_2 + r_3$

(C) r (D) $r_1r_2r_3 - 1$

45. If x + y = 2 is a tangent to the ellipse whose foci are (2, 3) and (3, 5), then

(A) length of minor axis is 6 (B) length of minor axis is 3

(C) length of major axis is $\frac{\sqrt{41}}{2}$ (D) eccentricity of ellipse is $\sqrt{\frac{5}{41}}$

46. If the lines joining the origin to the points of intersection of the line $3y = mx + 3$ and the curve $x^2 + y^2 = 1$ are at right angel then value of m + 3 is equal to

(A) 5 (B) 6

(C) 1 (D) Zero

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. If the length of intercept between coordinate axes, of common tangent in the first quadrant to the circle $x^2 + y^2 = 16$ and the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$ is L, then value of $3L^2$ = 190.

48. A tangent to the ellipse $x^2 + 4y^2 = 4$ meets the ellipse $x^2 + 2y^2 = 6$ at P and Q. If the angle between tangents at P and Q of ellipse $x^2 + 2y^2 = 6$ at angle α. Then value of $3\sin^2\alpha.\cos^2 2\alpha$ is

49. If a point on the parabola $y^2 = 4ax$. The normal at A cuts the parabola again at point B. If AB subtends a right angle at the vertex to parabola, If the slope of AB is m, then find value of m^2.

50. If $R = \left(15 + \sqrt{220}\right)^{19} + \left(15 + \sqrt{220}\right)^{82}$, then the digit at the unit place of [R] – 1 is……

51. If f(x) be a polynomial of degree 4 with leading coefficient 1 satisfies $f(1) = 10,\ f(2) = 20, f(3) = 30,$ then $f(12) + f(-8) - 19835$

52. Let $p = 144^{\sin^2 x} + 144^{\cos^2 x}$, then it the number of integral values of such p's can take, are m then value of m – 120 is.

53. If A, B and C are $n \times n$ matrices and $\det(A) = 2,\ \det(B) = 3$ and $\det(C) = 5$, then the value of $\frac{5}{2}\det\left(A^2BC^{-1}\right)$ is equal to

54. In a parabola $y = x^2$, vertex and two of its points are chosen to form a right triangle ABC with C as a right angle and vertex. If A is on 2nd quadrant and area of triangle is 1, then the sum of B's x-coordinates and y-coordinates is

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